KAIZEN MATHS

Lesson starter

Partial Fractions: Decomposition

Distinct Linear Factors

Name _____

S1

Express in partial fractions:

$\frac{-10x-28}{(x+1)(x+4)}$

S2

Express in partial fractions:

$\frac{-3x+21}{(x+3)(x-2)}$

S3

Express in partial fractions:

$\frac{5x-6}{(x-2)x}$

S4

Express in partial fractions:

$\frac{-3}{(x+1)(x+4)}$

Show your working

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Show your working

KAIZEN MATHS

Exit ticket

Partial Fractions: Decomposition

Distinct Linear Factors

Name _____

E1

Express in partial fractions:

$\frac{9x+39}{(x+5)(x+3)}$

E2

Express in partial fractions:

$\frac{2}{(x-3)(x-1)}$

E3

Express in partial fractions:

$\frac{-10x-4}{x(x+1)}$

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ALG-36C3-L1 · E

Show your working

Teacher answers & working

Partial Fractions: Decomposition · Distinct Linear Factors

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S1

Express in partial fractions:

$\frac{-10x-28}{(x+1)(x+4)}$

Answer:

$\frac{-6}{(x+1)} - \frac{4}{(x+4)}$

Working:

Step 1: Start with the correct form for two distinct linear factors.

$\frac{-10x-28}{(x+1)(x+4)} = \frac{A}{(x+1)} + \frac{B}{(x+4)}$

Step 2: Multiply through by (x + 1)(x + 4) to remove the denominators.

$-10x - 28 = A(x + 4) + B(x + 1)$

Step 3: Substitute x = -1 to eliminate B.

$-18 = A(3)$, so $A = -6$.

Step 4: Substitute x = -4 to eliminate A.

$12 = B(-3)$, so $B = -4$.

Step 5: Substitute A and B back into the partial fraction form.

Therefore $\frac{-10x-28}{(x+1)(x+4)} = \frac{-6}{(x+1)} - \frac{4}{(x+4)}$.

S2

Express in partial fractions:

$\frac{-3x+21}{(x+3)(x-2)}$

Answer:

$\frac{-6}{(x+3)} + \frac{3}{(x-2)}$

Working:

Step 1: Start with the correct form for two distinct linear factors.

$\frac{-3x+21}{(x+3)(x-2)} = \frac{A}{(x+3)} + \frac{B}{(x-2)}$

Step 2: Multiply through by (x + 3)(x - 2) to remove the denominators.

$-3x + 21 = A(x - 2) + B(x + 3)$

Step 3: Substitute x = -3 to eliminate B.

$30 = A(-5)$, so $A = -6$.

Step 4: Substitute x = 2 to eliminate A.

$15 = B(5)$, so $B = 3$.

Step 5: Substitute A and B back into the partial fraction form.

Therefore $\frac{-3x+21}{(x+3)(x-2)} = \frac{-6}{(x+3)} + \frac{3}{(x-2)}$.

Teacher answers & working

Partial Fractions: Decomposition · Distinct Linear Factors

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S3

Express in partial fractions:

$\frac{5x-6}{(x-2)x}$

Answer:

$\frac{2}{(x-2)} + \frac{3}{x}$

Working:

Step 1: Start with the correct form for two distinct linear factors.

$\frac{5x-6}{(x-2)x} = \frac{A}{(x-2)} + \frac{B}{x}$

Step 2: Multiply through by (x - 2)x to remove the denominators.

$5x - 6 = Ax + B(x - 2)$

Step 3: Substitute x = 2 to eliminate B.

$4 = A(2)$, so $A = 2$.

Step 4: Substitute x = 0 to eliminate A.

$-6 = B(-2)$, so $B = 3$.

Step 5: Substitute A and B back into the partial fraction form.

Therefore $\frac{5x-6}{(x-2)x} = \frac{2}{(x-2)} + \frac{3}{x}$.

S4

Express in partial fractions:

$\frac{-3}{(x+1)(x+4)}$

Answer:

$\frac{-1}{(x+1)} + \frac{1}{(x+4)}$

Working:

Step 1: Start with the correct form for two distinct linear factors.

$\frac{-3}{(x+1)(x+4)} = \frac{A}{(x+1)} + \frac{B}{(x+4)}$

Step 2: Multiply through by (x + 1)(x + 4) to remove the denominators.

$-3 = A(x + 4) + B(x + 1)$

Step 3: Substitute x = -1 to eliminate B.

$-3 = A(3)$, so $A = -1$.

Step 4: Substitute x = -4 to eliminate A.

$-3 = B(-3)$, so $B = 1$.

Step 5: Substitute A and B back into the partial fraction form.

Therefore $\frac{-3}{(x+1)(x+4)} = \frac{-1}{(x+1)} + \frac{1}{(x+4)}$.

Teacher answers & working

Partial Fractions: Decomposition · Distinct Linear Factors

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E1

Express in partial fractions:

$\frac{9x+39}{(x+5)(x+3)}$

Answer:

$\frac{3}{(x+5)} + \frac{6}{(x+3)}$

Working:

Step 1: Start with the correct form for two distinct linear factors.

$\frac{9x+39}{(x+5)(x+3)} = \frac{A}{(x+5)} + \frac{B}{(x+3)}$

Step 2: Multiply through by (x + 5)(x + 3) to remove the denominators.

$9x + 39 = A(x + 3) + B(x + 5)$

Step 3: Substitute x = -5 to eliminate B.

$-6 = A(-2)$, so $A = 3$.

Step 4: Substitute x = -3 to eliminate A.

$12 = B(2)$, so $B = 6$.

Step 5: Substitute A and B back into the partial fraction form.

Therefore $\frac{9x+39}{(x+5)(x+3)} = \frac{3}{(x+5)} + \frac{6}{(x+3)}$.

E2

Express in partial fractions:

$\frac{2}{(x-3)(x-1)}$

Answer:

$\frac{1}{(x-3)} - \frac{1}{(x-1)}$

Working:

Step 1: Start with the correct form for two distinct linear factors.

$\frac{2}{(x-3)(x-1)} = \frac{A}{(x-3)} + \frac{B}{(x-1)}$

Step 2: Multiply through by (x - 3)(x - 1) to remove the denominators.

$2 = A(x - 1) + B(x - 3)$

Step 3: Substitute x = 3 to eliminate B.

$2 = A(2)$, so $A = 1$.

Step 4: Substitute x = 1 to eliminate A.

$2 = B(-2)$, so $B = -1$.

Step 5: Substitute A and B back into the partial fraction form.

Therefore $\frac{2}{(x-3)(x-1)} = \frac{1}{(x-3)} - \frac{1}{(x-1)}$.

Teacher answers & working

Partial Fractions: Decomposition · Distinct Linear Factors

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E3

Express in partial fractions:

$\frac{-10x-4}{x(x+1)}$

Answer:

$\frac{-4}{x} - \frac{6}{(x+1)}$

Working:

Step 1: Start with the correct form for two distinct linear factors.

$\frac{-10x-4}{x(x+1)} = \frac{A}{x} + \frac{B}{(x+1)}$

Step 2: Multiply through by x(x + 1) to remove the denominators.

$-10x - 4 = A(x + 1) + Bx$

Step 3: Substitute x = 0 to eliminate B.

$-4 = A(1)$, so $A = -4$.

Step 4: Substitute x = -1 to eliminate A.

$6 = B(-1)$, so $B = -6$.

Step 5: Substitute A and B back into the partial fraction form.

Therefore $\frac{-10x-4}{x(x+1)} = \frac{-4}{x} - \frac{6}{(x+1)}$.